

The Entanglement First Law Tracks a Local Kernel; the Casini–Huerta–Myers Envelope Is Not Selected

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Abstract

Paper 25 found that the modular *hop* tracks the CHM envelope. Jacobson needs that envelope to do Clausius work: $\delta S = \delta \langle K_{\text{CHM}} \rangle$.

Two instruments were discarded before the gate was locked. Bond-local $\chi = \Delta S / \Delta t$ versus w fails the 1d theorem ($\rho \approx 0.28$): ΔC is nonlocal. A permutation of w is not a mutation: the intercept of $K_{ij} = aw_{ij} + b$ dominates.

The effective test: fit K_{CHM} on vacuum nearest neighbours, perturb every hop, and compare δS to $\text{Tr}(K_{\text{CHM}} \Delta C)$. Null: a *flat* local kernel, every hop equal to $\langle K_{\text{NN}} \rangle$. $\text{Tr}(K_{\text{exact}} \Delta C)$ is an identity and is not a gate.

1d $L = 24$: $\rho = 0.918$ (instrument PASS). Auditor $L = 20$: $\rho = 0.775$ (below the 0.80 lock; QUALIFIED). 3d, all 282 hops, $N = 12$, $R = 3$: $\rho = 0.850$, $R_{\text{shape}} = 0.527$. C1/C2 PASS. C3 FAIL: the flat kernel is better ($\rho = 0.912$, $R_{\text{shape}} = 0.411$). Auditor $N = 10$: same order (0.797 vs flat 0.865). C3 CONFIRMED.

The first law tracks a local kernel. The CHM envelope is not selected by Clausius. [P] as that negative. Not a refutation of Paper 25, and not Einstein.

1 Why these gates

A2 and Paper 25 are about the shape of K . The next path step is whether that shape is what generates δS . A test that a shapeless local K can win is the only mutation that can see the difference. It won.

2 Numbers

	ρ_{CHM}	R_{CHM}	ρ_{flat}	R_{flat}
1d interval	0.918	0.397	0.981	0.192
3d ball, 282 hops	0.850	0.527	0.912	0.411

Locality is the first-law input. The radial weight is extra structure this probe does not use. Combined with Papers 23–24, what actually feeds a Jacobson-style argument here is *local* K plus an area law, not a uniquely geometric envelope. Λ stays free. G is not measured.

Admission. *I designed the test so the CHM envelope had to beat a flat local kernel. It lost. I did not keep a permutation mutation after it proved vacuous, and I did not write that entanglement gravity is Einstein.*